
EMBEDDED HACKING

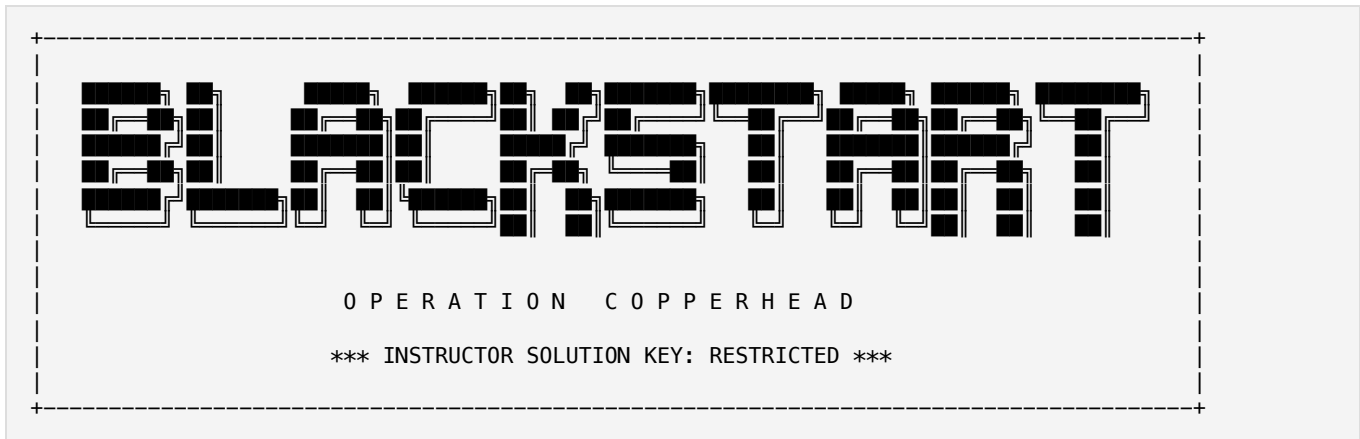
Reverse Engineering the Raspberry Pi Pico 2

Operation Copperhead - Instructor Solution Key

ARM Cortex-M33 • GDB • Ghidra

September 19, 2026

For educational and defensive security research only



The task and criterion headings in this key are word-for-word identical to `CTF-02-R.md`, so a student can match each criterion one-to-one.

Artifact Identity

Artifact	Value
Student image	<code>CTF-02.bin</code>
Flash image	<code>CTF-02.uf2</code>
Target	Raspberry Pi Pico 2 / RP2350 ARM Cortex-M33
Image base	<code>0x10000000</code>
Console	USB-CDC virtual COM, 115200 8N1

```
CTF-02.bin 85330C37CD0897746B1AF447E4BAC371DDE2042ABD2D61D58A61FE2A8EEF3537
CTF-02.uf2 F3CD4840260DB820D792758CECACC5297BEF1971B9EACF7601279256D8AF1EAB
```

Proof tool: `python3 scripts/verify_ctf.py` returns 26/26 checks passed against `CTF-02.bin`.

Task 1: Setup and Initial Analysis (12 points)

Solution

Criterion 1.1: Ghidra Project Setup (3 points). Import `CTF-02.bin` as Raw Binary, language ARM:LE:32:Cortex, base address `0x10000000`, then run auto-analysis.

Criterion 1.2: Vector Table Decoding (3 points). First 32 bytes of `CTF-02.bin`:

```
00 20 08 20 5B 01 00 10 1B 01 00 10 1D 01 00 10
11 01 00 10 11 01 00 10 11 01 00 10 11 01 00 10
```

Evidence	Answer
Vector table base	0x10000000
Initial SP	0x20082000
Reset pointer (as stored)	0x1000015B
Reset instruction address	0x1000015A

Criterion 1.3: main() and Status-Loop Addresses (4 points).

Element	Address
main()	0x100002E8
Recurring status loop start	0x1000034C
Loop back-edge (b.n 0x1000034C)	0x1000044E

Criterion 1.4: Thumb Addressing and Literal Pool (2 points). The stored reset pointer 0x1000015B has bit 0 set, selecting Thumb mode. Clearing bit 0 gives 0x1000015A. A representative literal pool entry is 0x100004B4, which holds 0x1000C4C8, the address of the format string "BLOCK STATE: %s", loaded by `ldr r0, [pc, #308]` at 0x1000037C.

Supporting Reference: SRAM Symbols (Ghidra names to semantic roles).

Ghidra Label	SRAM Address	Section	Role	Source Symbol
DAT_20001188	0x20001188	.data	Block length double	g_telemetry
DAT_20001198	0x20001198	.data	Signal key seed	g_auth_seed
DAT_2000119C	0x2000119C	.data	Track current	g_block_current
DAT_20001ECC	0x20001ECC	.bss	Dispatch state	g_dispatch_state
DAT_20001ED0	0x20001ED0	.bss	Fault poll counter	g_fault_polls
DAT_20001ED4	0x20001ED4	.bss	Input line buffer	g_linebuf
DAT_200020D4	0x200020D4	.bss	Input write index	g_lineidx
DAT_200020D8	0x200020D8	.bss	Operator state	g_operator_state
DAT_200020DC	0x200020DC	.bss	Derived signal key	g_signal_key

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 1.1: Ghidra Project Setup	3	Correct project name, ARM Cortex 32-bit little endian, base 0x10000000	One item off	Not set up
Criterion 1.2: Vector Table Decoding	3	Correct base, initial SP, reset pointer	One missing	Not found

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 1.3: main() and Status-Loop Addresses	4	Both addresses correct	One correct	Neither found
Criterion 1.4: Thumb Addressing and Literal Pool	2	Bit 0 cleared and one pool entry traced to its string	Partial	Incorrect

Instructor Notes & Assembly

- Confirm the Ghidra import used `Raw Binary`, `ARM:LE:32:Cortex`, base `0x10000000`, and that auto-analysis completed before any address was read.
- Verify `main()` is `0x100002E8`, the loop head is `0x1000034C`, and the back-edge is `0x1000044E`.
- For Criterion 1.4, accept any correctly traced literal pool entry; the pool entry `0x100004B4` holding `0x1000C4C8` (`"BLOCK STATE: %s"`, loaded at `0x1000037C`) is the reference example.
- The SRAM symbol table is supporting reference material, not a separate scored criterion.

Task 2: Find and Patch Bug #1: The Miscalibrated Release Threshold (15 points)

Solution

Criterion 2.1: Locate Compare Sites A and B (6 points).

```
10000302: 2b5e  cmp r3, #94  @ 0x5e
10000312: 2b5e  cmp r3, #94  @ 0x5e
```

Site	Address	File Offset	Original Bytes	Original Instruction
A	0x10000302	0x0302	5E 2B	cmp r3, #94
B	0x10000312	0x0312	5E 2B	cmp r3, #94

Criterion 2.2: Correct Immediate-Value Reasoning (4 points). The source constant is `SAFE_THRESHOLD = 95` and the test is `x < 95`. For an unsigned value, `x < 95` is exactly `x <= 94`, so the compiler emits `cmp r3, #94`. The correct limit is `60`, so the test is `x < 60`, which is `x <= 59`. The correct patched immediate is `0x3B (59)`, not `0x3C (60)`.

Criterion 2.3: Patch Compare Sites A and B (4 points).

Site	Address	File Offset	Original	Patched	After
A	0x10000302	0x0302	5E 2B	3B 2B	cmp r3, #59
B	0x10000312	0x0312	5E 2B	3B 2B	cmp r3, #59

Criterion 2.4: Explain Why Both Sites Must Be Patched (1 point). Site A drives the `BLOCK STATE` line and site B drives the `AUTO TRAIN` decision. Patching only site A makes the console read `CRITICAL` while the automated dispatch still says `AUTHORIZED`. Frozen reading `87 : 87 <= 94` is true (wrong); `87 <= 59` is false (correct).

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 2.1: Locate Compare Sites A and B	6	Both addresses and original bytes	One site	Not found
Criterion 2.2: Correct Immediate-Value Reasoning	4	Explains the <code><</code> to <code><=</code> transform and gives <code>0x3B</code>	Correct value, no reasoning	Wrong value
Criterion 2.3: Patch Compare Sites A and B	4	Both byte changes verified	One site	Not patched
Criterion 2.4: Explain Why Both Sites Must Be Patched	1	Clear explanation of the two independent comparisons	Vague	Missing

Instructor Notes & Assembly

- Both sites must be patched: `0x1000302` for `BLOCK STATE` and `0x1000312` for the `AUTO TRAIN` decision.
- The correct immediate is `0x3B` (59), not `0x3C` (60).
- Verify the byte changes on hardware; the corrected console reads `CRITICAL` and `HELD`.

Task 3: Find and Patch Bug #2: The False TRACK Banner (10 points)

Solution

Criterion 3.1: Locate the Banner String (3 points).

String	Address
"TRACK: NORMAL\r"	<code>0x1000C4B8</code>
"NORMAL" substring to patch	<code>0x1000C4BF</code>

The string is loaded in `main` and printed once at boot; it never reads the sensor.

Criterion 3.2: Patch Six Characters (4 points). `NORMAL` and `DANGER` are both six ASCII characters, so the patch preserves the length.

Address Range	Original Bytes	Patched Bytes
<code>0x1000C4BF</code> - <code>0x1000C4C4</code>	<code>4E 4F 52 4D 41 4C</code>	<code>44 41 4E 47 45 52</code>

Criterion 3.3: Character-by-Character Documentation (2 points).

Address	Original Char	Original Byte	Patched Char	Patched Byte
<code>0x1000C4BF</code>	N	<code>4E</code>	D	<code>44</code>
<code>0x1000C4C0</code>	O	<code>4F</code>	A	<code>41</code>
<code>0x1000C4C1</code>	R	<code>52</code>	N	<code>4E</code>
<code>0x1000C4C2</code>	M	<code>4D</code>	G	<code>47</code>

Address	Original Char	Original Byte	Patched Char	Patched Byte
0x1000C4C3	A	41	E	45
0x1000C4C4	L	4C	R	52

Criterion 3.4: Explain the Danger of a Hardcoded Status Word (1 point). The banner never consults the reading, so it reports a healthy track even while the frozen reading is dangerous, masking the hazard from the operator.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 3.1: Locate the Banner String	3	Correct address and cross-reference	Approximate	Not found
Criterion 3.2: Patch Six Characters	4	All six bytes changed, length preserved	Correct text, wrong bytes documented	Wrong length
Criterion 3.3: Character-by-Character Documentation	2	Original vs patched byte for all six characters	Partial	Missing
Criterion 3.4: Explain the Danger of a Hardcoded Status Word	1	Clear, specific reasoning	Generic	Missing

Instructor Notes & Assembly

- `NORMAL` and `DANGER` are both six characters; the patch must not change the string length or overwrite adjacent flash.
- Confirm the patch covers `0x1000C4BF` through `0x1000C4C4` exactly.
- The banner is printed once at boot and never recomputed, so it is a separate defect from the threshold.

Task 4: Find and Patch Bug #3: The Block Length Constant (10 points)

Solution

Criterion 4.1: Locate the .data Double (3 points). The console prints `BLOCK LENGTH: 3200 M` from the format string at `0x1000C4F0`. The value is a `double` in the `.data` init image at `0x1000EC60` (loaded into `0x20001188` at boot).

Criterion 4.2: IEEE-754 Bytes and Print Math (4 points). Eight bytes at `0x1000EC60`:

```

9A 99 99 99 99 99 09 40  -> 0x400999999999999A  -> 3.2 km  -> 3200 m
7B 14 AE 47 E1 7A D4 3F  -> 0x3FD47AE147AE147B  -> 0.32 km -> 320 m

```

Criterion 4.3: Patch to Print 320 M (3 points).

File Offset Range	Flash Address Range	Original Bytes	Patched Bytes
0xEC60 - 0xEC67	0x1000EC60 - 0x1000EC67	9A 99 99 99 99 99 09 40	7B 14 AE 47 E1 7A D4 3F

At `0x1000EC68` the adjacent telemetry fields (`03 00 00 00` flags and `07 00` crossing) are left untouched.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 4.1: Locate the .data Double	3	Correct address traced from the BLOCK LENGTH print	Approximate	Not found
Criterion 4.2: IEEE-754 Bytes and Print Math	4	Original and patched 8-byte double with print math (3200 M to 320 M)	Correct patch, no math	Wrong bytes
Criterion 4.3: Patch to Print 320 M	3	Console shows BLOCK LENGTH: 320 M	Wrong bytes	Not patched

Instructor Notes & Assembly

- The value is an 8-byte IEEE-754 double, not an integer. Follow the BLOCK LENGTH print into .data at `0x1000EC60`.
- The patched bytes `7B 14 AE 47 E1 7A D4 3F` decode to `0.32 km ( 320 m )`.
- Confirm the adjacent telemetry fields at `0x1000EC68` (`03 00 00 00` and `07 00`) are left untouched.

Task 5: Find and Patch Bug #4: The Signal Seed (15 points)

Solution

Criterion 5.1: Locate SIGNAL_SPEC and the Seed (5 points). The SIMPLE comparison in the loop is at `0x100003A6`:

```
100003a6: 4559    cmp r1, fp      ; fp = SIGNAL_SPEC = 0x2D879291 (pool 0x100004FC)
```

The corrupted seed is a `0x0A0A0A0A` word in the .data init image at `0x1000EC70` (loaded into `0x20001198` at boot). The byte values `0A 0A 0A 0A` also appear in the tbb jump table at `0x100003D4`; those are not the seed.

Criterion 5.2: Patch the Seed (4 points). The required seed is the ChaCha expand word `0x6B206574` ("te k"), stored little-endian as `74 65 20 6B`.

File Offset Range	Flash Address Range	Original Bytes	Patched Bytes
<code>0xEC70 - 0xEC73</code>	<code>0x1000EC70 - 0x1000EC73</code>	<code>0A 0A 0A 0A</code>	<code>74 65 20 6B</code>

Criterion 5.3: Explain the ARX Derivation (3 points). `derive_session_key(seed, iv)` works exactly like this:

- Set `a = seed`, `b = iv`, `c = 0x61707865`, `d = 0x3320646E`.
- Run four ChaCha quarter-rounds. A quarter-round runs four phases with rotate amounts 16, 12, 8, 7. Each phase is: `a = a + b`; `d = d XOR a`; `d = rotate_left(d, s)`; `c = c + d`; `b = b XOR c`; `b = rotate_left(b, s)`.
- Return `a XOR d`.

The runtime IV is `derive_session_key(0x6B206574, 0) = 0x43C974F6`. The shipped seed `0x0A0A0A0A` gives `derive_session_key(0x0A0A0A0A, 0x43C974F6) = 0x915DCFF8` (MISMATCH). The honest seed `0x6B206574` gives `derive_session_key(0x6B206574, 0x43C974F6) = 0x2D879291`, which equals `SIGNAL_SPEC` and prints `OK`.

Criterion 5.4: Separate the Security Layers (3 points). The signal seed only controls the local `SIGNAL KEY` telemetry check. It does not authenticate the operator. The Argon2id plus XChaCha20-Poly1305 gate is a separate layer that requires the 12-word passphrase.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 5.1: Locate <code>SIGNAL_SPEC</code> and the Seed	5	<code>0x2D879291</code> located and wrong seed <code>0x0A0A0A0A</code> found	Partial	Not found
Criterion 5.2: Patch the Seed	4	Seed bytes changed to <code>74 65 20 6B</code>	Wrong byte	Not patched
Criterion 5.3: Explain the ARX Derivation	3	Correct trace of the per-cycle derivation	Vague	Missing
Criterion 5.4: Separate the Security Layers	3	Correctly explains what the seed fixes versus the gate	Generic	Missing

Instructor Notes & Assembly

- The seed is at `0x1000EC70` in the `.data` init image. The `0A 0A 0A 0A` bytes in the `tbb` jump table at `0x100003D4` are not the seed.
- The patched seed `74 65 20 6B` is the little-endian form of the ChaCha expand word `0x6B206574` ("te k").
- The seed only affects the local `SIGNAL KEY` check; the Argon2id plus XChaCha20-Poly1305 gate is a separate authentication layer.

Task 6: GDB Register Capture of the Derived Key (15 points)

Solution

Criterion 6.1: Breakpoint at the Derive Return (4 points). The per-cycle derive is `bl derive_session_key` at `0x10000352`. Break at the next instruction, `0x10000356`, and read `$r0`. On the shipped image it is `0x915DCFF8`.

```
(gdb) break *0x10000356
(gdb) continue
(gdb) print/x $r0      # 0x915DCFF8 on the corrupted image
```

Criterion 6.2: Inspect the Two Arguments (4 points). At the call entry `0x10000352`:

Register	Corrupted image	Meaning
<code>\$r0</code>	<code>0x0A0A0A0A</code>	Seed
<code>\$r1</code>	<code>0x43C974F6</code>	Derived IV

Criterion 6.3: Override the Register (4 points). With execution at `0x10000356`, set `$r0` to the spec key, then continue. The next cycle prints `SIGNAL KEY: 0x2D879291 OK`.

```
(gdb) set $r0 = 0x2D879291
(gdb) continue
```

Criterion 6.4: Watchpoint on the Stored Key (3 points). The key is stored in SRAM at `0x200020DC` (`str r0, [r6, #0]`).

```
(gdb) watch *0x200020DC
```

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 6.1: Breakpoint at the Derive Return	4	Correct address and <code>\$r0</code> read as the bug-derived key	Address off	Not found
Criterion 6.2: Inspect the Two Arguments	4	Live seed and derived IV captured at the second call	One correct	Missing
Criterion 6.3: Override the Register	4	<code>\$r0</code> set to <code>0x2D879291</code> and the next cycle shows <code>OK</code>	Partial	Missing
Criterion 6.4: Watchpoint on the Stored Key	3	Watchpoint on the SRAM key location documented	Approximate	Missing

Instructor Notes & Assembly

- Confirm the breakpoint is placed at `0x10000356`, the instruction after the `bl derive_session_key` at `0x10000352`.
- On the shipped image `$r0` reads `0x915DCFF8`; `$r0` is the seed `0x0A0A0A0A` and `$r1` is the derived IV `0x43C974F6`.
- The stored key lives at `0x200020DC`; accept the documented watchpoint.

Task 7: Recover the Ouroboros Authority Frame (10 points)

Solution

Criterion 7.1: Locate Salt, Nonce, Ciphertext, and Tag (4 points).

Component	Flash Address	Size
Ciphertext + Tag	<code>0x1000CE94</code>	64 B
Nonce	<code>0x1000CED4</code>	24 B
Salt	<code>0x1000CEEC</code>	16 B

Criterion 7.2: Document Argon2id Parameters and Payload Contract (2 points). Argon2id: memory 64 KiB , iterations 3 , parallelism 1 , output key 32 B , salt the 16 bytes above. XChaCha20-Poly1305 decrypts the 48-byte ciphertext with the 16-byte tag. The plaintext is 01 68 65 6C 6C 6F 0D 0A followed by zeros: byte 0 turns the GPIO 25 LED on, and bytes 1 through 7 are printed as hello plus carriage-return and newline.

Criterion 7.3: Authenticate with the 12-Word Passphrase (2 points). At the RESPONSE> prompt, type:

```
orbit olive ladder marble quartz canyon ripple saddle violet ember walnut falcon
```

Expected result (proven on hardware):

```
hello
AUTHORITY FRAME: VERIFIED
```

Criterion 7.4: State the Honest Quantum Boundary (2 points). Grover-style search halves the effective security exponent of a symmetric key, so a 256-bit key gives about 128 bits of quantum security. The construction uses classical symmetric and password-hashing primitives and does not implement NIST post-quantum standards.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 7.1: Locate Salt, Nonce, Ciphertext, and Tag	4	All three addresses correct in flash	Two correct	Not found
Criterion 7.2: Document Argon2id Parameters and Payload Contract	2	Correct memory/time/parallelism and payload layout	Partial	Missing
Criterion 7.3: Authenticate with the 12-Word Passphrase	2	AUTHORITY FRAME: VERIFIED , LED on, payload printed	Partial	Not shown
Criterion 7.4: State the Honest Quantum Boundary	2	Grover halves symmetric exponents; not strict PQC	Generic	Misstates

Instructor Notes & Assembly

- Verify the three component addresses in flash: ciphertext plus tag at 0x1000CE94 , nonce at 0x1000CED4 , salt at 0x1000CEEC .
- The passphrase is fixed for the lab; a successful gate prints hello and AUTHORITY FRAME: VERIFIED and lights the on-board authority LED.
- Grade Criterion 7.4 on the honest boundary: 256-bit symmetric key maps to about 128 bits under Grover, and the design is not NIST post-quantum.

Task 8: Export and Verify (8 points)

Solution

Criterion 8.1: Export CTF-02_fixed.bin (1 point). Export the patched program from Ghidra (File -> Export Program... , Binary Format) as CTF-02_fixed.bin . The shipped image is 62,308 bytes.

Criterion 8.2: Convert to CTF-02_fixed.uf2 (1 point).

```
python uf2conv.py CTF-02_fixed.bin --base 0x10000000 --family 0xe48bff59 --output CTF-02_fixed.uf2
```

Criterion 8.3: Hardware Verification (4 points).

Before patching:

```
DEEPLINE METRO AUTHORITY
ADAPTIVE SIGNAL WINDOW: 38 MINUTES
USB-CDC 115200 8N1 | AUTHORIZED LAB CONSOLE
TRACK: NORMAL
BLOCK STATE: STABLE
AUTO TRAIN: AUTHORIZED
BLOCK LENGTH: 3200 M
FAULT POLLS: 1
SIGNAL KEY: 0x915DCFF8 MISMATCH
RESPONSE>
```

After all four patches:

```
DEEPLINE METRO AUTHORITY
ADAPTIVE SIGNAL WINDOW: 38 MINUTES
USB-CDC 115200 8N1 | AUTHORIZED LAB CONSOLE
TRACK: DANGER
BLOCK STATE: CRITICAL
AUTO TRAIN: HELD
BLOCK LENGTH: 320 M
FAULT POLLS: 1
SIGNAL KEY: 0x2D879291 OK
RESPONSE>
```

Criterion 8.4: Summary Table of All Patches (2 points).

#	Bug	File Offset	Flash Address	Original Bytes	Patched Bytes
1a	Operator threshold	0x0302	0x10000302	5E 2B	3B 2B
1b	Dispatch threshold	0x0312	0x10000312	5E 2B	3B 2B
2	TRACK banner	0xC4BF - 0xC4C4	0x1000C4BF - 0x1000C4C4	4E 4F 52 4D 41 4C	44 41 4E 47 45 52
3	Block length	0xEC60 - 0xEC67	0x1000EC60 - 0x1000EC67	9A 99 99 99 99 99 09 40	7B 14 AE 47 E1 7A D4 3F
4	Signal seed	0xEC70 - 0xEC73	0x1000EC70 - 0x1000EC73	0A 0A 0A 0A	74 65 20 6B

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 8.1: Export CTF-02_fixed.bin	1	Valid patched binary	Corrupted	Not submitted

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 8.2: Convert to CTF-02_fixed.uf2	1	Correct base and family flags	Wrong flags	Not submitted
Criterion 8.3: Hardware Verification	4	Corrected console output confirmed on hardware	Some lines corrected	No verification
Criterion 8.4: Summary Table of All Patches	2	Complete address and before/after table	Missing entries	No table

Instructor Notes & Assembly

- Verify the exported image with `python3 scripts/verify_ctf.py`; the shipped check expects 26/26 checks passed against CTF-02.bin.
- Confirm the UF2 conversion used base `0x10000000` and family `0xe48bff59`.
- The shipped image is 62,308 bytes; confirm the exported corrected image is a valid patched binary with all four fixes present.

Task 9: Written Reflection (5 points)

Solution

Criterion 9.1: "Rushed Build" Is Not an Excuse (2 points). The rebuild shipped four constants that were never checked against their documented limits, which is exactly what produced the false-safe reading. Pressure explains why the checks were skipped, not why they should be skipped.

Criterion 9.2: One Engineering Practice per Failure Area (3 points).

- Physical limits (Bugs #1 and #3): one shared configuration header plus a build-time assertion that each compiled limit matches its documented value.
- Banner (Bug #2): remove static banners; a hardware-in-the-loop test that compares displayed state to the live register.
- Seed integrity (Bug #4): reproducible builds with golden artifact hash comparison so keys and seeds match the certified specification.
- Image authenticity: enable RP2350 hardware secure boot with OTP hash verification so a modified image will not run.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 9.1: "Rushed Build" Is Not an Excuse	2	Specific, grounded reasoning	Generic	Missing
Criterion 9.2: One Engineering Practice per Failure Area	3	Concrete practices for the bugs and for image authenticity	Names some	Missing

Instructor Notes & Assembly

- Grade the specificity of the reasoning, not the length of the prose.

- Require concrete practices across the failure areas, including at least one practice for image authenticity.

How To Breadboard

- **Raspberry Pi Pico 2** powered over USB.
- **USB-CDC virtual serial console:** open the Pico's COM port at 115200 baud, 8 data bits, no parity, 1 stop bit.
- **SWD debug probe:** connect SWCLK, SWDIO, GND, and 3.3 V to the Pico debug header for GDB inspection and register capture.
- No other peripherals are required; the authority LED is on-board.

Complete Grading Summary

Task	Title	Points
Task 1	Setup and Initial Analysis	12
Task 2	Find and Patch Bug #1: The Miscalibrated Release Threshold	15
Task 3	Find and Patch Bug #2: The False TRACK Banner	10
Task 4	Find and Patch Bug #3: The Block Length Constant	10
Task 5	Find and Patch Bug #4: The Signal Seed	15
Task 6	GDB Register Capture of the Derived Key	15
Task 7	Recover the Ouroboros Authority Frame	10
Task 8	Export and Verify	8
Task 9	Written Reflection	5
TOTAL		100

Instructor Notes

Safety: Use only the supplied Pico 2, SWD probe, and firmware. Never connect the exercise to an operational railway, metro system, public network, military system, or third-party device.

Common Student Mistakes

- Patching only one threshold site (`0x10000302` or `0x10000312`), leaving one status line lying.
- Assuming the immediate equals the limit, producing an off-by-one boundary; the correct byte is `0x3B` (59), not `0x3C` (60).
- Replacing the banner string with a different length, corrupting adjacent flash; `NORMAL` and `DANGER` are both 6 bytes.
- Treating the block length as an integer and missing the 8-byte double in `.data`.
- Using the wrong `0.32` bytes and printing `316 M` instead of `320 M`.
- Treating the odd vector address `0x1000015B` as invalid instead of clearing bit 0 to get `0x1000015A`.

- Starting the seed patch at the wrong offset; the seed is at `0x1000EC70`.

Partial Credit Guidelines

- Award partial credit for one correct threshold site out of two, or for a correct immediate value without the `<` to `<=` reasoning.
- Award partial credit for a correct banner text with incorrectly documented bytes, or for partial character-by-character documentation.
- Award partial credit for a correct block-length patch without the IEEE-754 print math.
- Award partial credit for one of the two GDB argument captures, or for a partial register override.
- Award no credit for patches that change string length or overwrite adjacent flash.

Appendix: Expected Binary Diff

#	Bug	File Offset(s)	Flash Address(es)	Original Bytes	Patched Bytes
1a	Operator threshold	<code>0x0302</code>	<code>0x10000302</code>	<code>5E 2B</code>	<code>3B 2B</code>
1b	Dispatch threshold	<code>0x0312</code>	<code>0x10000312</code>	<code>5E 2B</code>	<code>3B 2B</code>
2	TRACK banner	<code>0xC4BF - 0xC4C4</code>	<code>0x1000C4BF - 0x1000C4C4</code>	<code>4E 4F 52 4D 41 4C</code>	<code>44 41 4E 47 45 52</code>
3	Block length	<code>0xEC60 - 0xEC67</code>	<code>0x1000EC60 - 0x1000EC67</code>	<code>9A 99 99 99 99 99 09 40</code>	<code>7B 14 AE 47 E1 7A D4 3F</code>
4	Signal seed	<code>0xEC70 - 0xEC73</code>	<code>0x1000EC70 - 0x1000EC73</code>	<code>0A 0A 0A 0A</code>	<code>74 65 20 6B</code>

Four defects, five changed regions: two immediate bytes (`0x3B 2B` at each threshold), six banner bytes, eight block-length bytes, and four seed bytes.